lab 12: Design and verification of up counter down counter and up-down counter.

Objectives

→ To design and implement up counter, down counter and up-down counter circuits using flipflops and basic digital logic components.

Theory

1) Up counter

→ An up counter is a digital circuit that increments its binary state with each incoming clock pluse. It typically starts from a value of zero and counts upward to its maximum capacity (2^n-1) before resetting back to zero.

fig: Block diagram

* Truth table

Clock pulse	Q_2	Q_1	Q_0
Initially(↑)	0	0	0
1 ↑	0	0	1
2 ↑	0	1	0
3 ↑	0	1	1
4 ↑	1	0	0
5 ↑	1	0	1
6 ↑	1	1	0
7 ↑	1	1	1
8 ↑	0	0	0
~~9 ↑~~			

2) Down Counter

→ A down counter performs the opposite function, decrementing its binary value by one for every clock cycle. It starts at a maximum present value and counts toward zero.

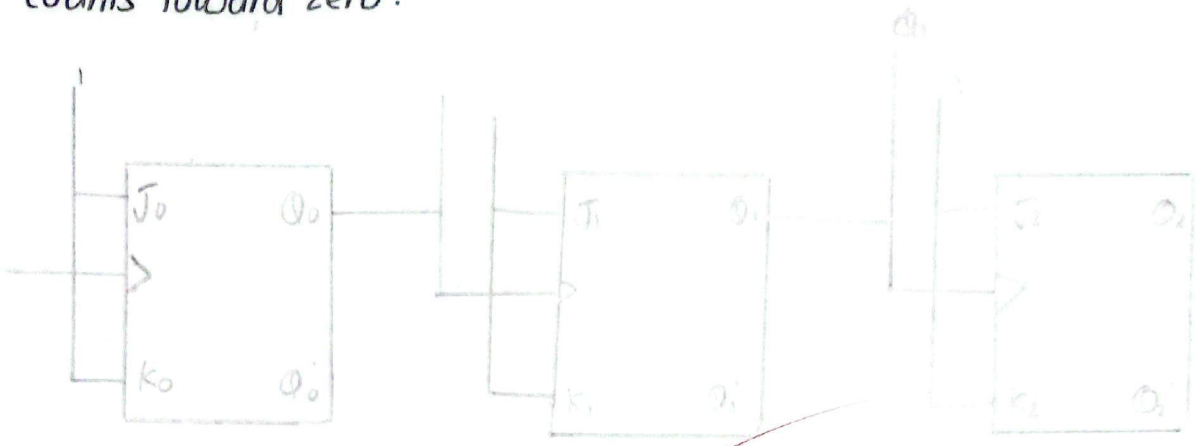


fig: Block diagram

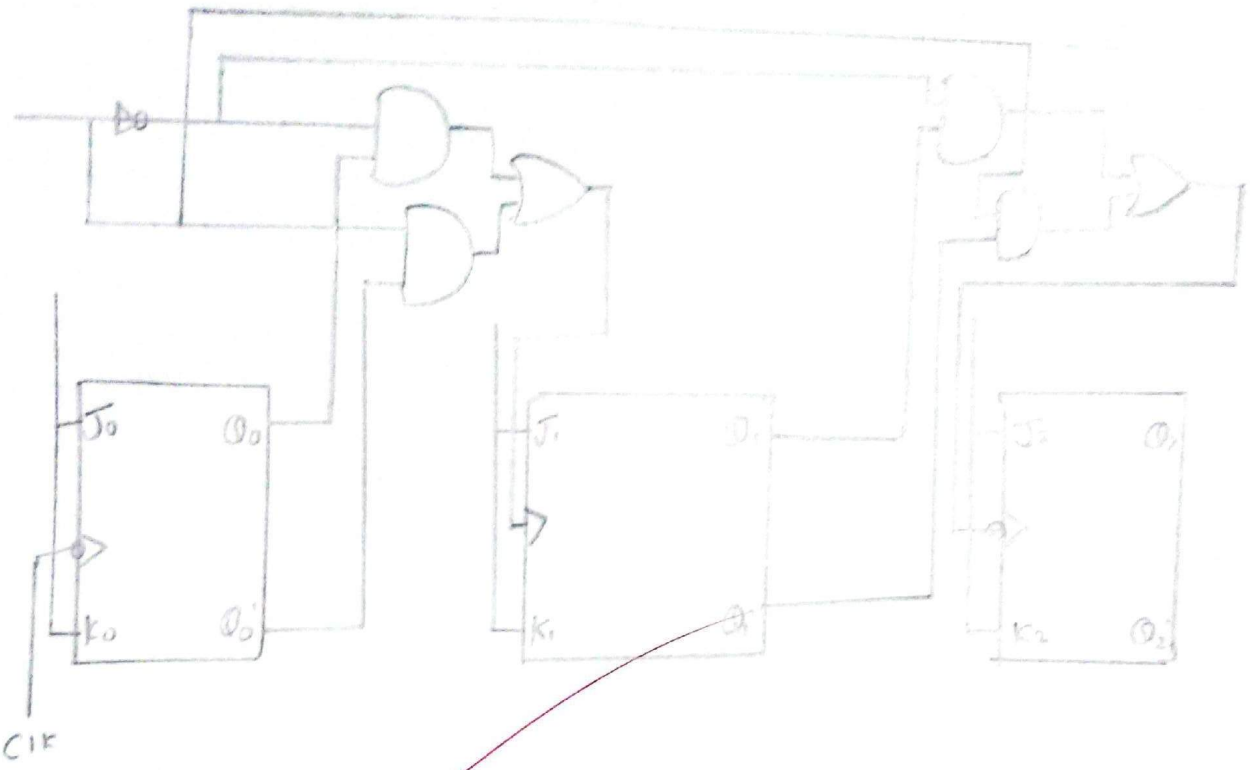
Truth table

Clock Pulse	Q_2	Q_1	Q_0
Initially ↑	1	1	1
1 ↑	1	1	0
2 ↑	1	0	1
3 ↑	1	0	0
4 ↑	0	1	1
5 ↑	0	1	0
6 ↑	0	0	1
7 ↑	0	0	0
8 ↑	1	1	1

3) Up-down counter

→ An up-down counter is a bidirectional circuit that can count in either direction based on the state of control input pin. This flexibility allows the system to switch between incrementing and decrementing modes without changing the underlying hardware architecture.

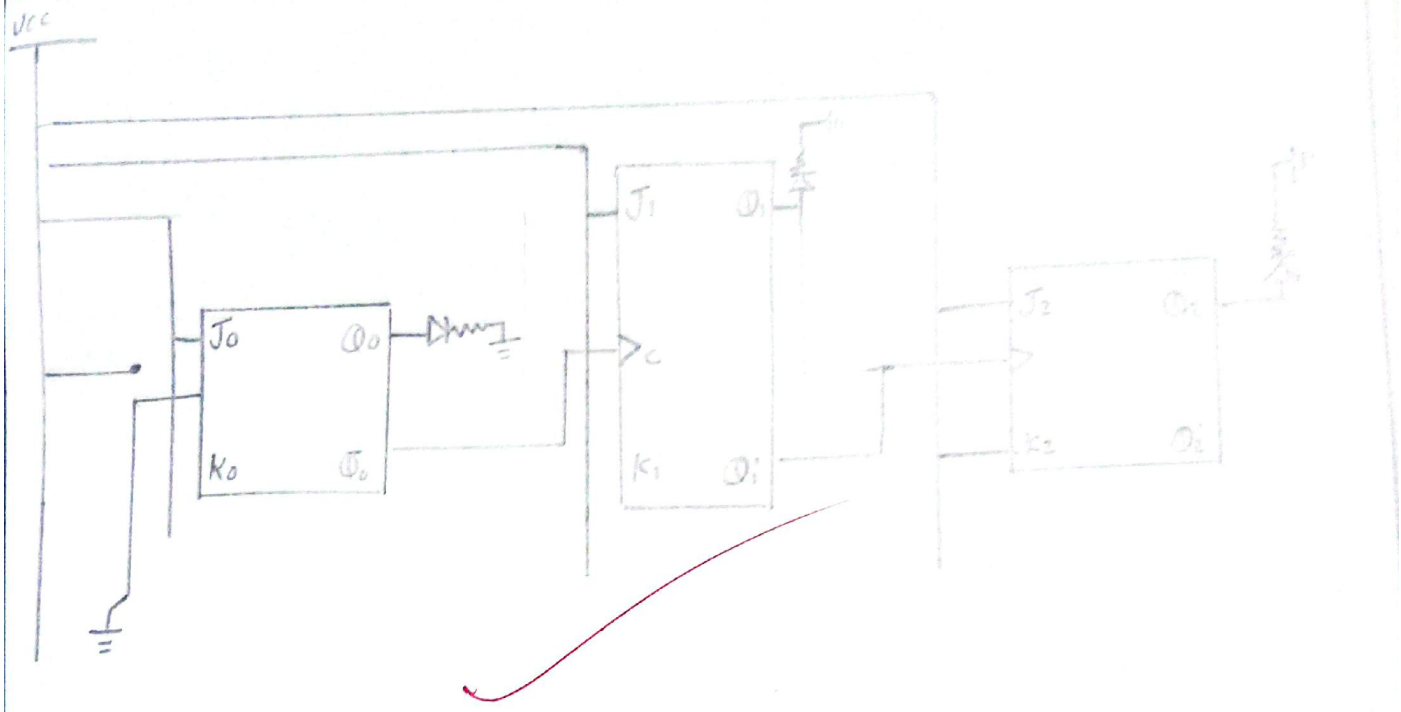
CamScanner

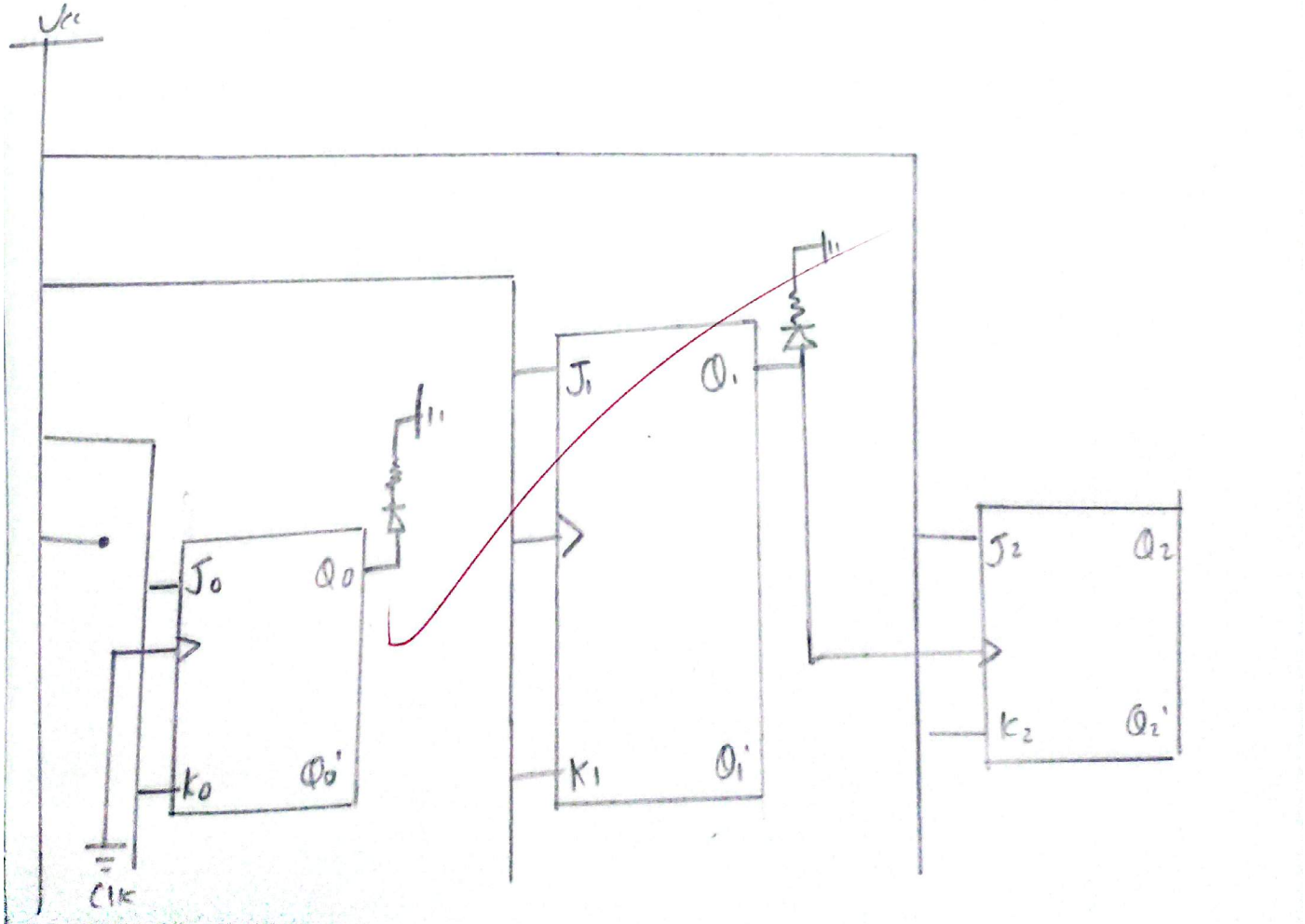


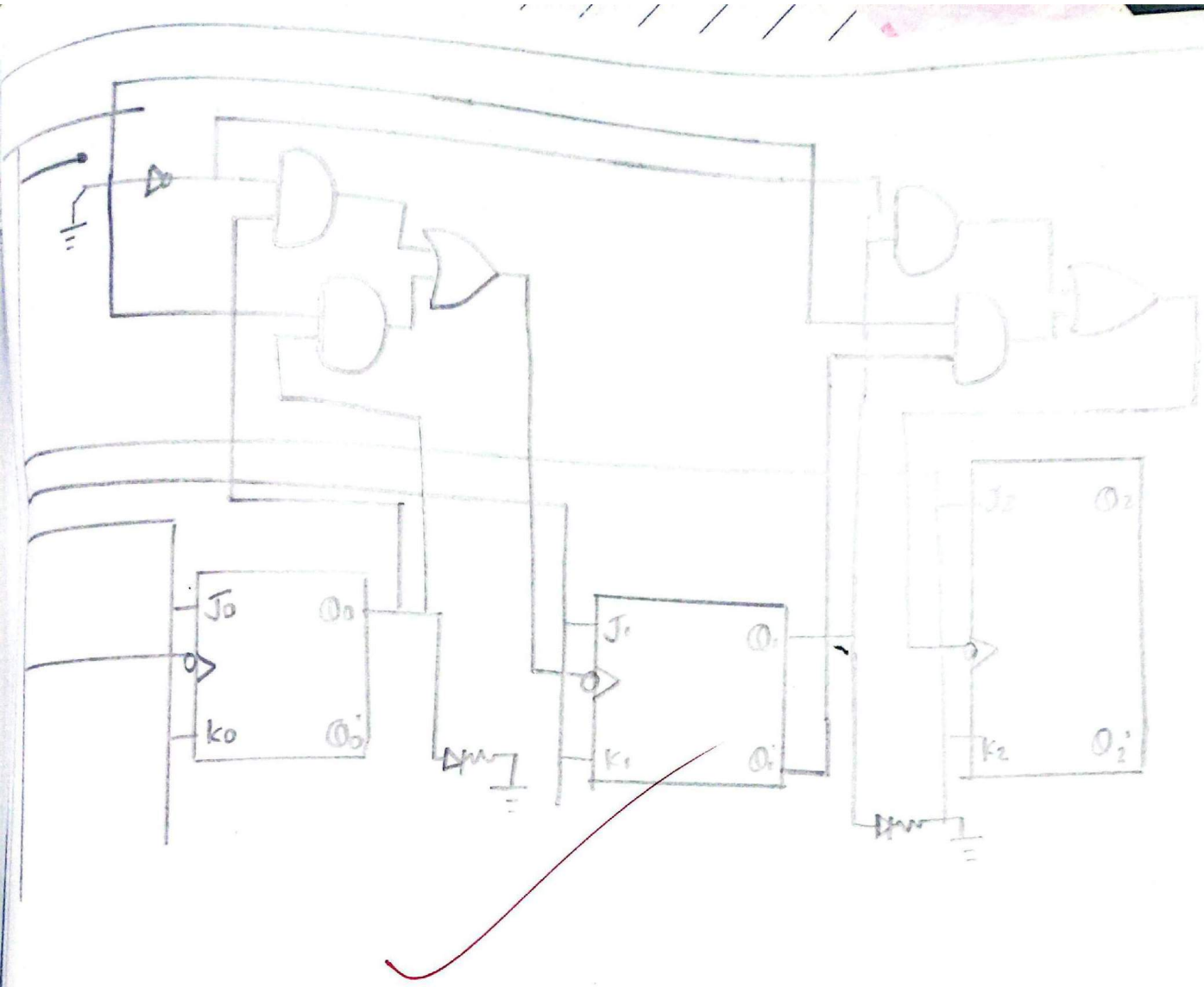
Truth table

	Current State				Next state		
	Q_2	Q_1	Q_0	Q_0	Q_2	Q_1	Q_0
1	0	0	0		0	0	1
1	0	0	1		0	1	0
1	0	1	0		0	1	1
1	0	1	1		1	0	0
1	1	0	0		1	0	1
1	1	0	1		1	1	0
1	1	1	0		1	1	1
1	1	1	1		0	0	0
0	0	0	0		1	1	0
0	0	0	1		1	0	1
0	0	1	0		1	0	0
0	0	1	1		0	1	1
0	1	0	0		0	1	0
0	1	0	1		0	0	1
0	1	1	0		0	0	0
0	1	1	1		1	1	1

Observation







Conclusion

In this laboratory session we designed and complemented up counter, down counter.